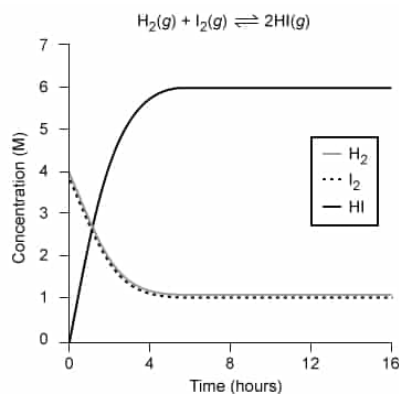




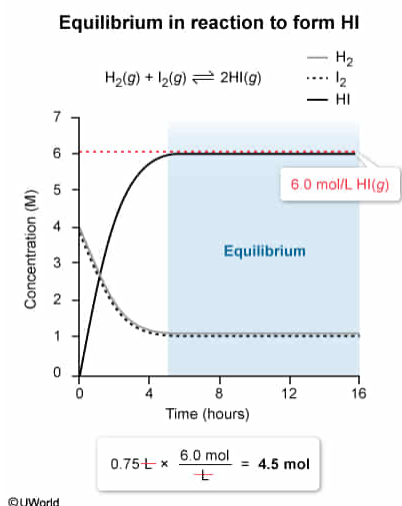
Equilibrium is reached in a reversible reaction when the rate of the forward reaction equals the rate of the reverse reaction. The graph shows the changes in concentration during the formation of $\text{HI}(\text{g})$ from a mixture of $\text{H}_2(\text{g})$ and $\text{I}_2(\text{g})$. If the measurement is taken at equilibrium, how many moles of $\text{HI}(\text{g})$ are present in a 750 mL sample of the reaction mixture?

- ☐ A. 0.8 mol [6%]
☐ B. 1.9 mol [34%]
☒ C. 4.5 mol [48%]
☐ D. 6.0 mol [10%]

Correct

47% answered correctly

Explanation:



Equilibrium is achieved in a reversible reaction when the forward reaction and the reverse reaction occur simultaneously at the **same rate**. At **equilibrium**, the forward reaction generates products as fast as the reverse reaction converts those products back into the original reactants, and this causes the **concentrations** of the reactants and the products to become **constant**. Although constant, the equilibrium concentrations are not necessarily equal because *equilibrium* refers to a state of equal reaction *rates* (ie, changes in concentration over time) but not to a state of equal concentrations.

In the graph of concentration vs. time for the reaction between $\text{H}_2(\text{g})$ and $\text{I}_2(\text{g})$, the constant (horizontal) regions of the curves (after approximately 5 hours) indicate a state of equilibrium in which the concentrations of the chemical species no longer change. From this region of the graph, the equilibrium concentration of $\text{HI}(\text{g})$ is seen to be 6.0 mol/L. Therefore, if a 750 mL sample of the reaction mixture were analyzed at equilibrium, the number of moles of $\text{HI}(\text{g})$ present in the sample would be determined as follows:

750

mL

×

(Choice A) At equilibrium, a 750 mL sample of the reaction mixture contains 0.8 mol each of $\text{H}_2(\text{g})$ and $\text{I}_2(\text{g})$, but the question asks for the moles of $\text{HI}(\text{g})$.

(Choice B) A 750 mL sample measured at the intersection point of the lines on the graph (after approximately 1 hour) contains 1.9 moles of $\text{HI}(\text{g})$, but this point does not indicate the equilibrium of the reaction. This is the point at which the concentrations of the reactants and products are equal, but equilibrium refers to equal *rates*, not equal concentrations.

(Choice D) At equilibrium, the $\text{HI}(g)$ has a molar *concentration* of 6.0 mol/L . This is the number of moles of $\text{HI}(g)$ per liter (ie, *molarity*) rather than the number of *moles* in a 750 mL sample of the reaction mixture.

Educational objective:

Equilibrium is achieved when two opposing chemical reactions occur simultaneously at the same rate such that the concentrations of the chemical species become constant. Equilibrium reaction rates are equal, but the equilibrium concentrations of chemical species may be unequal.

Time spent: 0 sec

Subject:
General Chemistry

QID: 402325

System:
Thermodynamics, Kinetics & Gas Laws